ZENTROMART

VOICE AI SUPPORT AGENT

VOICEOPS 2026 CAPSTONE | FROM TOKENS TO TRANSFORMERS

IIT Guwahati × Gnani.ai

A voice-first customer support prototype combining speech recognition, knowledge retrieval, conversational handling, mock business tools, and speech synthesis.

01 | EXECUTIVE SUMMARY

ZentroMart is a prototype for voice-based customer support. It accepts customer audio, transcribes speech, retrieves relevant passages from a supplied knowledge base, routes requests through a conversation agent, and generates a spoken response. Mock business tools support order lookup and ticket creation where appropriate.

VOICE INTERFACE	KNOWLEDGE GROUNDED	BUSINESS ACTIONS
Gnani.ai STT converts customer speech to text; Gnani.ai TTS produces an audio reply.	Sentence Transformers and FAISS retrieve relevant knowledge-base passages for response grounding.	Order lookup requires verification; ticket creation depends on the mock tool's returned status.

02 | SYSTEM WORKFLOW

1. Audio Input	2. STT	3. Agent	4. RAG	5. Tools	6. TTS
Customer recording	Speech → text	Intent & dialogue	FAISS retrieval	Verify / ticket	Text → speech

The pipeline saves the transcript, generated text response, synthesized audio, retrieved-source metadata, and run metadata in the output directory.

03 | IMPLEMENTATION

COMPONENT	ROLE IN THE PROTOTYPE
Voice pipeline	Coordinates audio input, STT, conversation processing, TTS, WAV-header repair, and output artifact saving.
RAG pipeline	Chunks knowledge-base content, creates embeddings with Sentence Transformers, and builds/uses a FAISS similarity index.
Conversation agent	Processes customer requests, retrieves relevant context, and routes supported requests to mock business tools.
Mock tools	Provides order lookup with identity verification and ticket creation through the supplied tool interface.
Baseline / evaluation	Includes a baseline script and evaluation output for the supplied question set.

04 | SAFETY & RELIABILITY

IDENTITY VERIFICATION

Protected order information should only be disclosed after the supplied verification flow succeeds. Follow the tool's retry rules.

POLICY GROUNDING

Use current knowledge-base policy documents and exclude superseded material from retrieval.

TOOL CONFIRMATION

Do not claim that a ticket was created unless the mock tool reports success.

SECRETS & DATA

Keep API credentials in a local environment file; do not commit secrets or private customer data.

05 | TECHNOLOGY STACK

Python	**Gnani.ai** STT / TTS	**Sentence Transformers**	**FAISS**	**Pandas / NumPy**	**Google GenAI**

06 | TESTING & EVALUATION STATUS

Local validation confirmed that the principal Python modules compiled without syntax errors and that a sample voice-pipeline run completed successfully. The generated artifacts include a transcript, a text reply, a synthesized WAV reply, retrieved-source metadata, and run metadata.

CHECK	OBSERVATION	INTERPRETATION
Syntax compilation	Completed successfully for run.py and principal pipeline modules.	No syntax errors were reported by py_compile.
Sample voice run	Pipeline returned success and saved output artifacts.	Demonstrates a working local audio-processing path.
Mock ticket tool	A ticket-creation test returned success.	Confirms one successful tool invocation, not broad test coverage.
Baseline evaluation	The 52-row CSV contains baseline answers with Gemini API quota errors (429 RESOURCE_EXHAUSTED).	Valid baseline accuracy cannot be inferred from this run; scores are intentionally not claimed.

Evaluation integrity note
The available baseline output records a quota failure rather than completed model answers. Therefore, no baseline score or comparative improvement score is reported here. A valid Step 1 vs Step 2 comparison requires successful evaluation outputs from both systems.

07 | SAMPLE OUTPUT ARTIFACTS

FILE	PURPOSE
customer_question_transcript.txt	Text recognized from the sample audio input.
customer_question_reply.txt	Customer-facing text response generated by the pipeline.
customer_question_reply.wav	Synthesized spoken response.
customer_question_sources.json	Retrieved source passages and related metadata.
voice_pipeline_results.json	Run status and pipeline metadata.

08 | LIMITATIONS

The current implementation is a capstone prototype and depends on external speech services, API availability, local dataset configuration, and mock business tools. The evaluation CSV currently records Gemini quota failures for the baseline answers, so comparative performance remains unverified. The sample voice run establishes a working path but does not prove production-level robustness or accuracy.

09 | FUTURE IMPROVEMENTS

01	**Complete reproducible evaluation** Run baseline and RAG evaluations successfully; report grounded-answer accuracy, retrieval relevance, and tool-handling results.
02	**Improve robustness** Add retries and clear fallbacks for API limits, unavailable services, malformed audio, and missing knowledge-base content.
03	**Strengthen safety testing** Test identity-verification failures, retry limits, protected-data disclosure, policy conflicts, and ticket-creation failures.
04	**Measure voice experience** Measure latency, transcription quality, response usefulness, and audio quality over a larger test set.

10 | REPRODUCIBILITY & PROJECT LINKS

Source code and setup instructions: github.com/dk-khandelwal06/ZentroMart-VoiceAI
Install dependencies with `python -m pip install -r requirements.txt` and run the local pipeline with `python run.py` after configuring the supplied dataset path and required API credentials.

PREPARED BY

Daksh Khandelwal

B.S. in Applied Artificial Intelligence & Data Science

Indian Institute of Technology Jodhpur (IIT Jodhpur)

dk.khandelwaliitj@gmail.com | GitHub | LinkedIn